

ECE180J Advanced Renewable Energy Sources

Lecture 10: Probability Theory Revisited

Dr. Yu Zhang

ECE Department

University of California, Santa Cruz

- H. Stark and J. Woods, "Probability, Random Processes, and Estimation Theory for Engineers".
- A. Papoulis, "Probability, Random Variables, and Stochastic Processes".

Probability Space

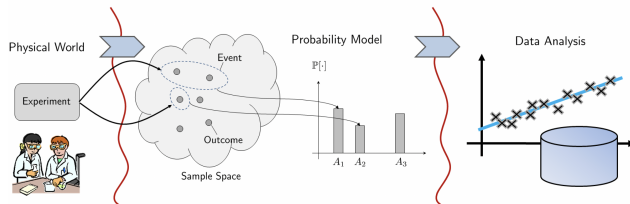


Figure: Fig source: Prof. Stanley Chan at Purdue Univ.

Probability space is a three-tuple $(\Omega, \mathcal{F}, P)$:

- **Sample space** Ω : the set of all outcomes of a random experiment.
 - Ω may be finite ($\{H, T\}$), countably infinite ($\Omega = \{1, 2, 3, \dots\}$), or uncountably infinite ($[0, 1]$).

Probability Space

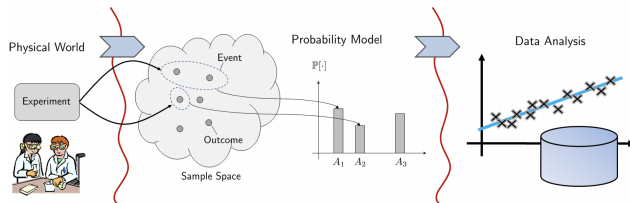


Figure: Fig source: Prof. Stanley Chan at Purdue Univ.

Probability space is a three-tuple $(\Omega, \mathcal{F}, P)$:

- **Sample space** Ω : the set of all outcomes of a random experiment.
 - Ω may be finite ($\{H, T\}$), countably infinite ($\Omega = \{1, 2, 3, \dots\}$), or uncountably infinite ($[0, 1]$).
- **Event space** $\mathcal{F}$: a set whose elements $A \in \mathcal{F}$ are subsets of Ω .

Probability Space

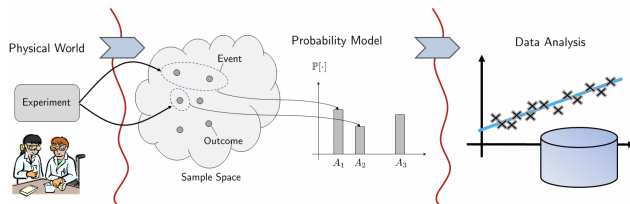


Figure: Fig source: Prof. Stanley Chan at Purdue Univ.

Probability space is a three-tuple $(\Omega, \mathcal{F}, P)$:

- **Sample space** Ω : the set of all outcomes of a random experiment.
 - Ω may be finite ($\{H, T\}$), countably infinite ($\Omega = \{1, 2, 3, \dots\}$), or uncountably infinite ($[0, 1]$).
- **Event space** $\mathcal{F}$: a set whose elements $A \in \mathcal{F}$ are subsets of Ω .
- **Probability function** P : satisfies three axioms:
 - $P(A) \geq 0, \forall A \in \mathcal{F}$.
 - $P(\Omega) = 1$.
 - If $A_1, A_2, \dots$ are mutually exclusive events ($A_i \cap A_j = \emptyset$, for $i \neq j$), then $P(\bigcup_i A_i) = \sum_i P(A_i)$.

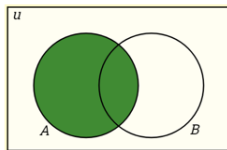
Properties of Probability

Consider events $A, B, C \subseteq \Omega$. Let $P(AB)$ denote $P(A \cap B)$.

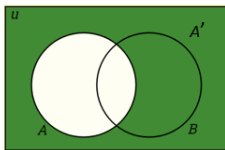
- $A \subseteq B \Rightarrow P(A) \leq P(B)$.
- $P(AB) \leq \min\{P(A), P(B)\}$.
- $P(A \cup B) = P(A) + P(B) - P(AB)$.
- $P(\bar{A}) = 1 - P(A)$: $\bar{A} = \Omega \setminus A$.
- Complement rule (De Morgan's Law): $\overline{AB} = \bar{A} \cup \bar{B}$; $\overline{A \cup B} = \bar{A}\bar{B}$
- $P(AB) = P(A|B)P(B) = P(B|A)P(A)$.
- **Conditional probability:** If $P(B) > 0$, then $P(A|B) = \frac{P(AB)}{P(B)} = \frac{P(B|A)P(A)}{P(B)}$ (Bayes' theorem).
- If A and B are independent, then $P(AB) = P(A)P(B)$.
- Distributive law: $(A \cup B) \cap C = (AC) \cup (BC)$.

Set Operations and Venn Diagrams

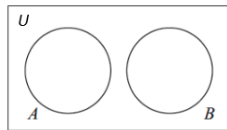
Set Operations and Venn Diagrams



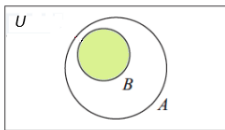
Set A



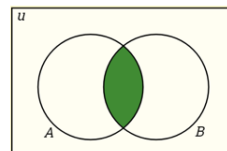
A' the complement of A



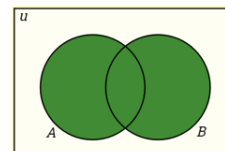
A and B are disjoint sets



B is proper subset of A
 $B \subset A$



Both A and B



Either A or B

Random Variables

Definition (Random Variable (RV))

Let Ω be the sample space of an experiment, and $\mathbb{R}$ denote the set of real numbers. Then, a *random variable* $X : \Omega \mapsto \mathbb{R}$ associated with this experiment is a function that assigns each outcome in Ω to a real number. The range of X is denoted as $\text{val}(X)$.

Random Variables

Definition (Random Variable (RV))

Let Ω be the sample space of an experiment, and $\mathbb{R}$ denote the set of real numbers. Then, a *random variable* $X : \Omega \mapsto \mathbb{R}$ associated with this experiment is a function that assigns each outcome in Ω to a real number. The range of X is denoted as $\text{val}(X)$.

Example. Flip a coin 5 times. Let X denote the random variable for the number of times the coin came up heads. Then $X(\omega_0) = 3$, for the outcome $\omega_0 = \{\text{HHTHT}\}$.

There are two different types of random variables that are often studied: *discrete* and *continuous*.

Random Variables (Cont'd)

If X is a discrete random variable, we use the notation

$$\Pr(X = k) := \Pr(\{\omega : X(\omega) = k\})$$

for the probability of the event $X = k$.

If X is a continuous random variable, we use the notation

$$\Pr(a \leq X \leq b) := \Pr(\{\omega : a \leq X(\omega) \leq b\})$$

for the probability of the event $a \leq X \leq b$.

Cumulative Distribution Function (CDF)

Definition

Let X be a random variable associated with an experiment. Then,

$$F_X(x) := \Pr(X \leq x)$$

is a *cumulative distribution function*.

Properties of CDFs:

- (a) $0 \leq F_X(x) \leq 1$.
- (b) $\lim_{x \rightarrow -\infty} F_X(x) = 0$ and $\lim_{x \rightarrow +\infty} F_X(x) = 1$.
- (c) F_X is nondecreasing, namely if $x \leq y$ then $F_X(x) \leq F_X(y)$.
- (d) F_X is right-continuous, i.e. $\lim_{x \rightarrow a^+} F_X(x) = F_X(a)$.

CDF for Discrete and Continuous RVs

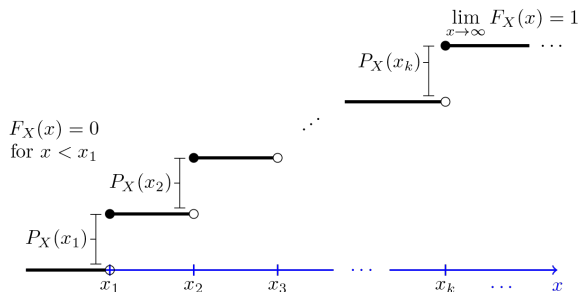


Figure: CDF of a discrete random variable.

CDF for Discrete and Continuous RVs

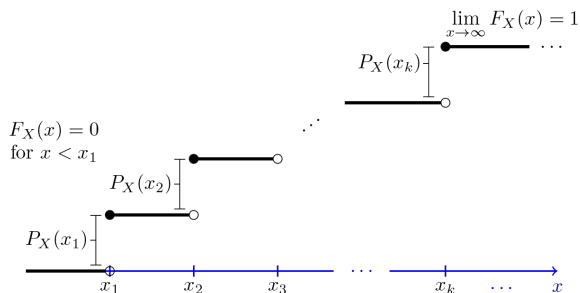


Figure: CDF of a discrete random variable.

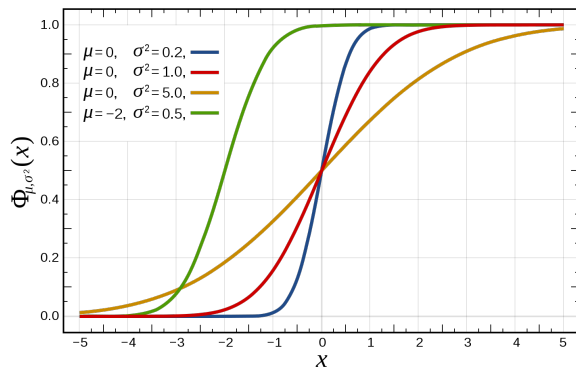


Figure: CDF of a continuous random variable.

PMF and PDF

For discrete random variables, we have the *probability mass function* (PMF):

$$p_X(x) := \Pr(X = x),$$

where $\sum_{x \in \text{val}(X)} p_X(x) = 1$.

¹Note that F_X may not be everywhere differentiable even for continuous random variable X .

PMF and PDF

For discrete random variables, we have the *probability mass function* (PMF):

$$p_X(x) := \Pr(X = x),$$

where $\sum_{x \in \text{val}(X)} p_X(x) = 1$.

For continuous random variables, we instead consider *probability density function* (PDF),

$$f_X(x) := \frac{dF_X(x)}{dx} = F'_X(x),$$

provided that F_X is differentiable at x .¹ Notice that $f_X(x)$ and $\Pr(X = x)$ are two different concepts, which can be related by

$$\Pr(x \leq X \leq x + \Delta x) \approx f_X(x)\Delta x$$

and

$$\Pr(X \in A) = \int_{x \in A} f_X(x) dx,$$

where $A \subseteq \text{val}(X)$.

¹Note that F_X may not be everywhere differentiable even for continuous random variable X .

Common Distributions

Name of the probability distribution	Probability distribution function	Mean	Variance
Binomial distribution	$\Pr(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}$	np	$np(1 - p)$
Geometric distribution	$\Pr(X = k) = (1 - p)^{k-1} p$	$\frac{1}{p}$	$\frac{(1 - p)}{p^2}$
Normal distribution	$f(x \mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$	μ	σ^2
Uniform distribution (continuous)	$f(x a, b) = \begin{cases} \frac{1}{b-a} & \text{for } a \leq x \leq b, \\ 0 & \text{for } x < a \text{ or } x > b \end{cases}$	$\frac{a + b}{2}$	$\frac{(b - a)^2}{12}$
Exponential distribution	$f(x \lambda) = \lambda e^{-\lambda x}$	$\frac{1}{\lambda}$	$\frac{1}{\lambda^2}$

Figure: PDF or PMF of commonly used random variables (Wiki).

Gaussian (Normal) Distribution

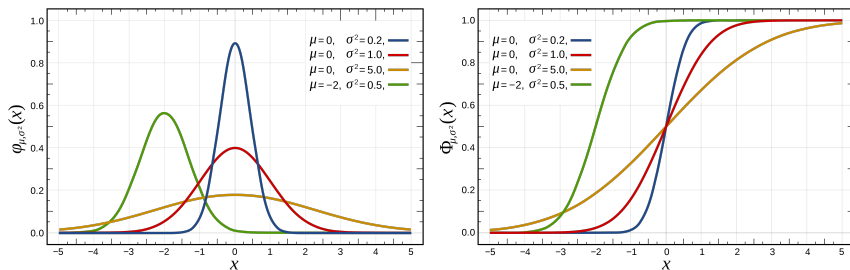


Figure: PDF and CDF of Gaussian distribution.

PDF of Gaussian distribution:

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$$

CDF of standard Gaussian distribution ($\mu = 0$ and $\sigma = 1$):

$$\Phi(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^x e^{-t^2/2} dt$$

Expectation

Definition (Expectation)

Let $g : \mathbb{R} \rightarrow \mathbb{R}$ be a function and X be a random variable, discrete or continuous, then the *expectation* of the random variable $g(X)$ is

$$\mathbb{E}[g(X)] := \sum_{x \in \text{val}(X)} g(x)p_X(x) \quad \text{or} \quad \mathbb{E}[g(X)] := \int_{x \in \text{val}(X)} g(x)f_X(x)dx,$$

respectively.

Linearity of the expectation operator:

- (a) $\mathbb{E}[ag(X) + bh(X)] = a\mathbb{E}[g(X)] + b\mathbb{E}[h(X)]$ for any constants a, b , and arbitrary functions $g(\cdot), h(\cdot)$.
- (b) $X \perp\!\!\!\perp Y \Rightarrow \mathbb{E}[XY] = \mathbb{E}[X]\mathbb{E}[Y]$.

Expectation

Definition (Expectation)

Let $g : \mathbb{R} \rightarrow \mathbb{R}$ be a function and X be a random variable, discrete or continuous, then the *expectation* of the random variable $g(X)$ is

$$\mathbb{E}[g(X)] := \sum_{x \in \text{val}(X)} g(x)p_X(x) \quad \text{or} \quad \mathbb{E}[g(X)] := \int_{x \in \text{val}(X)} g(x)f_X(x)dx,$$

respectively.

Linearity of the expectation operator:

- (a) $\mathbb{E}[ag(X) + bh(X)] = a\mathbb{E}[g(X)] + b\mathbb{E}[h(X)]$ for any constants a, b , and arbitrary functions $g(\cdot), h(\cdot)$.
- (b) $X \perp\!\!\!\perp Y \Rightarrow \mathbb{E}[XY] = \mathbb{E}[X]\mathbb{E}[Y]$.

The *indicator function*: $\mathbf{1}_A := \begin{cases} 1, & \text{if } A \text{ is true} \\ 0, & \text{otherwise.} \end{cases} \Rightarrow \mathbb{E}[\mathbf{1}_A] = \Pr(A) \times 1 + \Pr(\bar{A}) \times 0 = \Pr(A),$

$$F_X(x) = \mathbb{E}[\mathbf{1}_{\{X \leq x\}}].$$

Variance

Given a distribution of a random variable, its variance to measure how concentrated that distribution is around the expectation (mean). Formally, we have

Definition (Variance)

$\text{Var}[X] := \mathbb{E}[(X - \mathbb{E}[X])^2] = \mathbb{E}[X^2] - (\mathbb{E}[X])^2$, where $\mathbb{E}[X^2]$ is the second moment of X .

The following can be derived immediately from the definition:

- (a) $\text{Var}[cX] = c^2 \text{Var}[X]$.
- (b) $\text{Var}[c] = 0$ for any constant c .
- (c) $\text{Var}[aX \pm bY] = a^2 \text{Var}[X] + b^2 \text{Var}[Y] \pm 2ab \times \text{Cov}[X, Y]$.

Examples

Let $X \sim \exp(\lambda)$ whose density function is $f_X(x) = \lambda e^{-\lambda x}$. Find $E[X]$ and $\text{Var}[X]$.

Examples

Let $X \sim \exp(\lambda)$ whose density function is $f_X(x) = \lambda e^{-\lambda x}$. Find $E[X]$ and $\text{Var}[X]$. **Solution:** From the definition of expectation and integration by parts, we have

$$\begin{aligned} E(X) &= \int_0^{\infty} x f_X(x) dx \\ &= \lambda \int_0^{\infty} x e^{-\lambda x} dx \\ &= -x e^{-\lambda x} \Big|_0^{\infty} + \int_0^{\infty} e^{-\lambda x} dx \\ &= 0 + \frac{e^{-\lambda x}}{-\lambda} \Big|_0^{\infty} = \frac{1}{\lambda} . \end{aligned} \qquad \begin{aligned} V(X) &= \int_0^{\infty} x^2 f_X(x) dx - \frac{1}{\lambda^2} \\ &= \lambda \int_0^{\infty} x^2 e^{-\lambda x} dx - \frac{1}{\lambda^2} \\ &= -x^2 e^{-\lambda x} \Big|_0^{\infty} + 2 \int_0^{\infty} x e^{-\lambda x} dx - \frac{1}{\lambda^2} \\ &= -x^2 e^{-\lambda x} \Big|_0^{\infty} - \frac{2x e^{-\lambda x}}{\lambda} \Big|_0^{\infty} - \frac{2}{\lambda^2} e^{-\lambda x} \Big|_0^{\infty} - \frac{1}{\lambda^2} = \frac{2}{\lambda^2} - \frac{1}{\lambda^2} = \frac{1}{\lambda^2} . \end{aligned}$$

Multivariate Random Variables

Definition (Joint CDF)

Let X, Y be two random variables associated with an experiment. Then the *joint cumulative distribution function* of X and Y is

$$F_{XY} := \Pr(X \leq x, Y \leq y)$$

and the *marginal cumulative distribution function* of X is

$$F_X(x) := \lim_{y \rightarrow +\infty} \Pr(X \leq x, Y \leq y).$$

Multivariate Random Variables

Definition (Joint CDF)

Let X, Y be two random variables associated with an experiment. Then the *joint cumulative distribution function* of X and Y is

$$F_{XY} := \Pr(X \leq x, Y \leq y)$$

and the *marginal cumulative distribution function* of X is

$$F_X(x) := \lim_{y \rightarrow +\infty} \Pr(X \leq x, Y \leq y).$$

Similarly, the *joint PMF* $p_{XY}(x, y) := \Pr(X = x, Y = y)$

the *marginal PMF* of X , $p_X(x) := \sum_{y \in \text{val}(Y)} \Pr(X = x, Y = y)$.

The *joint PDF* of continuous X and Y is $f_{XY}(x, y) := \frac{\partial^2 F_{XY}(x, y)}{\partial x \partial y}$

the *marginal PDF* of X : $f_X(x) := \int_{y \in \text{val}(Y)} f_{XY}(x, y) dy$.

Conditional Distribution

The above summation or integral operation is called *marginalization*. Note that

$$\iint_A f_{XY}(x, y) dx dy = \Pr((x, y) \in A).$$

Definition (Conditional Distribution for Discrete RVs)

Let X and Y be discrete random variables. The *conditional distribution* of Y given $X = x$ is

$$p_{Y|X}(y | x) := \frac{p_{XY}(x, y)}{p_X(x)},$$

provided that $p_X(x) \neq 0$.

We say that X and Y are *independent* if $p_{Y|X}(y | x) = p_Y(y)$.

Note that $X \perp\!\!\!\perp Y \Leftrightarrow p_{XY}(x, y) = p_X(x)p_Y(y)$.

Conditional Distribution

The above summation or integral operation is called *marginalization*. Note that

$$\iint_A f_{XY}(x, y) dx dy = \Pr((x, y) \in A).$$

Definition (Conditional Distribution for Discrete RVs)

Let X and Y be discrete random variables. The *conditional distribution* of Y given $X = x$ is

$$p_{Y|X}(y | x) := \frac{p_{XY}(x, y)}{p_X(x)},$$

provided that $p_X(x) \neq 0$.

We say that X and Y are *independent* if $p_{Y|X}(y | x) = p_Y(y)$.

Note that $X \perp\!\!\!\perp Y \Leftrightarrow p_{XY}(x, y) = p_X(x)p_Y(y)$.

Q: If $X \perp\!\!\!\perp Y$. For arbitrary functions $g(\cdot)$ and $h(\cdot)$, are $g(X) \perp\!\!\!\perp h(Y)$?

A: Yes.

Covariance & Correlation

Definition (Covariance and Correlation)

Let X and Y be two random variables. Their *covariance* is defined as

$$\text{Cov}[X, Y] := E[(X - E[X])(Y - E[Y])] = E[XY] - E[X]E[Y].$$

Their *correlation* is defined as

$$\text{Corr}[X, Y] := \frac{\text{Cov}[X, Y]}{\sqrt{\text{Var}(X)\text{Var}(Y)}}.$$

Covariance & Correlation

Definition (Covariance and Correlation)

Let X and Y be two random variables. Their *covariance* is defined as

$$\text{Cov}[X, Y] := E[(X - E[X])(Y - E[Y])] = E[XY] - E[X]E[Y].$$

Their *correlation* is defined as

$$\text{Corr}[X, Y] := \frac{\text{Cov}[X, Y]}{\sqrt{\text{Var}(X)\text{Var}(Y)}}.$$

1. $\text{Corr}[X, Y] \in [-1, 1]$ is a measure of **linear association** between X and Y .
2. X and Y are uncorrelated if $\text{Corr}[X, Y] = 0$.
3. $\text{Corr}[X, Y] = \pm 1 \Leftrightarrow Y = aX + b$ for some constants a and b .
4. If $X \perp\!\!\!\perp Y \implies$ they are uncorrelated.
5. X and Y can be uncorrelated yet dependent due to a nonlinear relationship.

Example: $X \sim N(0, 1), Y = X^2 \implies \text{Corr}[X, Y] = E[XY] - E[X]E[Y] = E[X^3] = 0$ (all odd-order moments of X are equal to zero). Hence, X and Y are uncorrelated. But they're clearly dependent.

Covariance: Example 1

Let X and Y be discrete random variables, with joint probability function $p_{X,Y}$ given by

$$p_{X,Y}(x, y) = \begin{cases} 1/2 & x = 3, y = 4 \\ 1/3 & x = 3, y = 6 \\ 1/6 & x = 5, y = 6 \\ 0 & \text{otherwise.} \end{cases}$$

Then $E(X) = (3)(1/2) + (3)(1/3) + (5)(1/6) = 10/3$, and $E(Y) = (4)(1/2) + (6)(1/3) + (6)(1/6) = 5$. Hence,

$$\begin{aligned} \text{Cov}(X, Y) &= E((X - 10/3)(Y - 5)) \\ &= (3 - 10/3)(4 - 5)/2 + (3 - 10/3)(6 - 5)/3 + (5 - 10/3)(6 - 5)/6 \\ &= 1/3. \blacksquare \end{aligned}$$

Covariance: Example 2

Let X be any random variable with $\text{Var}(X) > 0$. Let $Y = 3X$, and let $Z = -4X$. Then $\mu_Y = 3\mu_X$ and $\mu_Z = -4\mu_X$. Hence,

$$\begin{aligned}\text{Cov}(X, Y) &= E((X - \mu_X)(Y - \mu_Y)) = E((X - \mu_X)(3X - 3\mu_X)) \\ &= 3 E((X - \mu_X)^2) = 3 \text{Var}(X),\end{aligned}$$

while

$$\begin{aligned}\text{Cov}(X, Z) &= E((X - \mu_X)(Z - \mu_Z)) = E((X - \mu_X)((-4)X - (-4)\mu_X)) \\ &= (-4)E((X - \mu_X)^2) = -4 \text{Var}(X).\end{aligned}$$

Note in particular that $\text{Cov}(X, Y) > 0$, while $\text{Cov}(X, Z) < 0$. Intuitively, this says that Y increases when X increases, whereas Z decreases when X increases. ■

PDF of Sum of Two Random Variables

For independent random variables X and Y , the distribution f_Z of $Z = X + Y$ equals the **convolution** of f_X and f_Y :

$$f_Z(z) = \int_{-\infty}^{\infty} f_Y(z - x) f_X(x) dx$$

Theorem (Law of Large Numbers (LLN))

Let $X_1, X_2, \dots, X_n$ be a sequence of independent and identically distributed (i.i.d.) random variables so that $E[X_1] = E[X_2] = \dots = E[X_n] < \infty$. Let

$$\bar{X}_n := \frac{X_1 + X_2 + \dots + X_n}{n}$$

denote the sample mean of those n random variables. Then $\bar{X}_n \rightarrow E[X_1]$ as $n \rightarrow \infty$ almost surely (a.s., strong law) and in probability (i.p., weak law).

Theorem (Central Limit Theorem (CLT))

Let $X_1, X_2, \dots, X_n$ be a sequence of i.i.d. random variables, and assume that $E[X_i] = \mu$ and $\text{Var}[X_i] = \sigma^2$, for all i . Let $S_n := X_1 + X_2 + \dots + X_n$. Then, $E[S_n] = n\mu$, $\text{Var}[S_n] = n\sigma^2$ and we have the standardization of S_n ,

$$\frac{S_n - n\mu}{\sigma\sqrt{n}} \xrightarrow{i.p.} N(0, 1) \text{ as } n \rightarrow \infty$$

where $N(0, 1)$ denotes the standard normal random variable.